

Algebra

Home Work 2

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Question 1.3.1

Let us put the permutations in cycle form:

$$\sigma = (1\ 3\ 5)(2\ 4)$$

$$\tau = (1\ 5)(2\ 3)$$

Then:

$$\sigma^2 = [(1\ 3\ 5)(2\ 4)][(1\ 3\ 5)(2\ 4)] = (1\ 5\ 3)$$

$$\sigma\tau = [(1\ 3\ 5)(2\ 4)][(1\ 5)(2\ 3)] = (1)(2\ 5\ 3\ 4) = (2\ 5\ 3\ 4)$$

$$\tau\sigma = [(1\ 5)(2\ 3)][(1\ 3\ 5)(2\ 4)] = (1\ 2\ 4\ 3)$$

$$\tau^2\sigma = \sigma = (1\ 3\ 5)(2\ 4) \quad (\text{Because } \tau^2 = 1)$$

Any distinct cycles of order 2 when squared become 1.

Question 1.3.2

Let us first create the two permutations in cycle form:

$$\begin{aligned}\sigma &= (1\ 13\ 5\ 10)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9) \\ \tau &= (1\ 14)(2\ 9\ 15\ 13\ 4)(3\ 10)(5\ 12\ 7)(8\ 11)\end{aligned}$$

Then:

$$\begin{aligned}\sigma^2 &= [(1\ 13\ 5\ 10)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9)][(1\ 13\ 5\ 10)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9)] \\ &= (1\ 5)(3\ 8\ 15)(4\ 11\ 12)(7\ 9\ 14)(10\ 13)\end{aligned}$$

$$\begin{aligned}\sigma\tau &= [(1\ 13\ 5\ 10)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9)][(1\ 14)(2\ 9\ 15\ 13\ 4)(3\ 10)(5\ 12\ 7)(8\ 11)] \\ &= (1\ 11\ 3)(2\ 4)(5\ 9\ 8\ 7\ 10\ 15)(13\ 14)\end{aligned}$$

$$\tau\sigma = [(1\ 14)(2\ 9\ 15\ 13\ 4)(3\ 10)(5\ 12\ 7)(8\ 11)][(1\ 13\ 5\ 10)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9)]$$

$$= (1\ 4)(2\ 9)(3\ 13\ 12\ 15\ 11\ 5)(8\ 10\ 14)$$

$$\begin{aligned}\tau^2\sigma &= [(2\ 15\ 4\ 9\ 13)(5\ 7\ 12)][(1\ 13\ 5\ 10)(3\ 15\ 8)(4\ 14\ 11\ 7\ 12\ 9)] \quad (1.3.2.1) \\ &= (1\ 2\ 15\ 8\ 3\ 4\ 14\ 11\ 12\ 13\ 7\ 5\ 10)\end{aligned}$$

Where in (1.3.2.1) I have used the fact that distinct cycles of order 2 when squared give 1. Hence:

$$\tau^2 = [(2\ 9\ 15\ 13\ 4)(5\ 12\ 7)][(2\ 9\ 15\ 13\ 4)(5\ 12\ 7)] = (2\ 15\ 4\ 9\ 13)(5\ 7\ 12)$$

Question 1.6.3

(a) Assume that $\psi : G \rightarrow H$ is an isomorphism.

Let G be an abelian group and $h_1, h_2 \in H$ be any two elements in H . Then, as ψ is a bijection, $\exists g_1, g_2 \in G$ such that: $h_1 = \psi(g_1)$; $h_2 = \psi(g_2)$. Then:

$$h_1 h_2 = \psi(g_1) \psi(g_2) = \psi(g_1 g_2) = \psi(g_2 g_1) = \psi(g_2) \psi(g_1) = h_2 h_1$$

Consequently H must also be abelian. So, when $G \cong H$ we must have that G abelian $\implies H$ abelian.

ASIDE:

As ψ is an isomorphism (i.e. bijection) we have that $\psi^{-1} : H \rightarrow G$ must exist. We now show that ψ^{-1} must also be an isomorphism. To do this let $h_1, h_2 \in H$ be any two elements in H .

Then, as before, there $\exists g_1, g_2 \in G$ such that: $h_1 = \psi(g_1)$; $h_2 = \psi(g_2)$ OR $\psi^{-1}(h_1) = g_1$; $\psi^{-1}(h_2) = g_2$. Then:

$$\begin{aligned} & \psi(g_1 g_2) = \psi(g_1) \psi(g_2) && \text{(as } \psi \text{ is an isomorphism)} \\ \implies & \psi^{-1}(\psi(g_1 g_2)) = \psi^{-1}(\psi(g_1) \psi(g_2)) \\ & \implies g_1 g_2 = \psi^{-1}(h_1 h_2) \\ \implies & \psi^{-1}(h_1) \psi^{-1}(h_2) = \psi^{-1}(h_1 h_2) && \text{(i.e. } \psi^{-1} \text{ is also an isomorphism)} \end{aligned}$$

□

If we now assume that H is abelian we can use $\psi^{-1} : H \rightarrow G$ - which is also an isomorphism - and the same logic as above to conclude that G must also be abelian.

Conclusion: $\psi : G \rightarrow H$ is an isomorphism $\implies (G \text{ is abelian} \implies H \text{ is abelian})$.

(b) In the case where $\psi : G \rightarrow H$ is a homomorphism then H will be abelian when G is abelian if the map ψ is **surjective**.

This is seen from the very first proof that " G abelian implies H abelian". Here we needed to ensure that any two elements of H had corresponding elements in G .

Question 1.6.9

Prove that D_{24} and S_4 are not isomorphic.

Both groups have 24 elements - so to prove they are not isomorphic we need to find a structure present in one and not in the other.

The group S_4 has only the following elements that are of order 2:

$$(1\ 2), (1\ 3), (1\ 4), (2\ 3), (2\ 4), (3\ 4), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)$$

of which there are 9.

D_{24} has at least 12 different reflections which are each of order 2. This difference in structure means that these two groups can't be isomorphic to each other.

Question 2.1.2

(a) The set of 2-cycles in S_n (for $n \geq 3$) is NOT a subgroup of S_n .

Take $(1\ 2)$ and $(2\ 3)$ then:

$$(1\ 2)(2\ 3) = (1\ 2\ 3)$$

which is NOT a 2-cycle. As composition is not stable (i.e. closed) on this subset of S_n (for $n \geq 3$) it can't be a subgroup.

(b) The set of reflections in D_{2n} (for $n \geq 3$) is NOT a subgroup of D_{2n} .

There are n such reflections. Each time we perform a reflection the "clockwise" (or "anti-clockwise") ordering of the nodes is reversed. If we take two distinct reflections and compose them we know that what we will obtain no inversion of the node ordering AND we won't have the identity mapping. No single reflection can do this - hence this set is not stable under composition and so is NOT a subgroup of D_{2n} (for $n \geq 3$).

(c) Given $n > 1$ and composite. Given there exists a $g \in G$, where G is a group, and $|g| = n$. Then the set: $\{x \in G \mid |x| = n\} \cup \{1\}$ is NOT a subgroup of G .

Let x be an element of this set. Then $x^2, x^3, \dots, x^{n-1}$ are also elements of this set if this set is a group. Now $n = kp$, where $1 < k, p < n$, hence this sequence must also contain x^k . However $|x^k| = p < n$. This means that this element (x^k) is NOT in this set and hence the set is NOT a group - it is not stable (i.e. closed).

(d) The set of (positive and negative) odd integers in $\mathbb{Z}$ together with 0 is NOT a group.

The reason that this set is NOT a group is because the sum of any two odd integers is even and hence this set is not stable (i.e. closed) under addition.

(e) The set of reals whose square is a rational number (under addition) is NOT a group.

We can show this using a counter example. Both $\sqrt{2}$ and $\sqrt{3}$ belong to this set. But $\sqrt{2} + \sqrt{3}$ is NOT in this set since its square is $5 + 2\sqrt{6}$ - which is irrational. This set is therefore NOT a group as it is not stable (i.e. closed) under addition.

Question 2.1.6

Let G be an abelian group and $K = \{g \in G \mid |g| < \infty\}$

Let us check that it has the properties needed to be a subgroup of G :

- **Subset:** Clearly: $K \subseteq G$ - i.e. K is a subset of G .
- **Identity:** As $1^1 = 1 \Rightarrow 1 \in K$ and K is not empty!
- **Inverse:** If $k \in K \Rightarrow k^n = 1$ for some $n \in \mathbb{N}$. Then:

$$k^n = 1 \implies (k^{-n})k^n = k^{-n}1 \implies 1 = k^{-n} = [k^{-1}]^n \quad (2.1.6.1)$$

The RHS of (2.1.6.1) shows that: $(k^{-1})^n = 1$ and hence $k^{-1} \in K$

- **Closure:** Let $a, b \in K \implies \exists m, n \in \mathbb{N}$ such that: $a^m = 1$ and $b^n = 1$. Then:

$$(ab)^{mn} = a^{mn}b^{mn} = (a^m)^n(b^n)^m = (1)^n(1)^m = 1 \quad (2.1.6.2)$$

where, in (2.1.6.2), we used the abelian property to rearrange all the a 's and b 's to come together. Hence, K is also closed.

We have shown that K is a subgroup of G .

Now we consider an example for a non-abelian group G , for which K is NOT a subgroup.

Question 2.1.8

Let H and K both be subgroups of G . Assume that $H \cup K \leq G$ (i.e. $H \cup K$ is a subgroup of G).

Now take any $a \in H$ & $b \in K$, then we must have that: $(a + b) \in H \cup K$, since: $a, b \in H \cup K$ and $H \cup K \leq G$. Consequently:

$$(a + b) \in H \cup K \implies (a + b) \in H \text{ or } (a + b) \in K$$

Assume that $(a + b) \in H$. Then we must have that:

$$\begin{aligned} a \in H &\implies (-a) \in H && \text{(since H is a subgroup of G)} \\ &\implies (a + b) + (-a) \in H && \text{(as both are elements of H)} \\ &\implies b \in H \end{aligned}$$

Since true for any $b \in K$ we must conclude that: $K \subseteq H$.

This argument is symmetric and if we had assumed that $(a + b) \in K$ we would have been led to: $H \subseteq K$.

□

Question 2.3.3

The generators of $Z/48Z$ will be all those elements: $\bar{k}$ such that $(k, 48) = 1$. As $48 = 3 \cdot 2^4$ we see that the $\bar{k}$ values we want are:

$$\bar{k} = 1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47$$

Question 2.3.13

In this section we just need to find one property that is NOT shared between the groups to prove that they are not isomorphic.

(a) The group $\mathbf{Z} \times \mathbf{Z}_2$ has a non-zero element which is its own inverse - i.e. $(0, 1)$, since $(0, 1) + (0, 1) = (0, 0)$. The group $\mathbf{Z}$ has no non-zero element which is its own inverse.

Consequently, these two groups can't be isomorphic: i.e. $\mathbf{Z} \times \mathbf{Z}_2 \not\cong \mathbf{Z}$

(b) The groups $\mathbf{Q} \times \mathbf{Z}_2$ and $\mathbf{Q}$ are NOT isomorphic.

The proof is exactly the same as was used in part (a) and hence: $\mathbf{Q} \times \mathbf{Z}_2 \not\cong \mathbf{Q}$

Question 2.3.14

We are given that: $\sigma = (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12)$ - which is a cycle of length 12.
From this we conclude that: $\sigma^{12} = 1$.

Hence:

$$\begin{aligned}
 \sigma^{13} &= \sigma \cdot \sigma^{12} &= \sigma &= (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12) \\
 \sigma^{65} &= \sigma^5 \cdot (\sigma^{12})^5 &= \sigma^5 &= (1\ 6\ 11\ 4\ 9\ 2\ 7\ 12\ 5\ 10\ 3\ 8) \\
 \sigma^{626} &= \sigma^2 \cdot (\sigma^{12})^{52} &= \sigma^2 &= (1\ 3\ 5\ 7\ 9\ 11)(2\ 4\ 6\ 8\ 10\ 12) \\
 \sigma^{1195} &= \sigma^7 \cdot (\sigma^{12})^{99} &= \sigma^7 &= (1\ 8\ 3\ 10\ 5\ 12\ 7\ 2\ 9\ 4\ 11\ 6) \\
 \sigma^{-6} &= \sigma^6 \cdot (\sigma^{12})^{-1} &= \sigma^6 &= (1\ 7)(2\ 8)(3\ 9)(4\ 10)(5\ 11)(6\ 12) \\
 \sigma^{-81} &= \sigma^3 \cdot (\sigma^{12})^7 &= \sigma^3 &= (1\ 4\ 7\ 10)(2\ 5\ 8\ 11)(3\ 6\ 9\ 12) \\
 \sigma^{-570} &= \sigma^6 \cdot (\sigma^{12})^{-48} &= \sigma^6 &= (1\ 7)(2\ 8)(3\ 9)(4\ 10)(5\ 11)(6\ 12) \\
 \sigma^{-1211} &= \sigma \cdot (\sigma^{12})^{-101} &= \sigma &= (1\ 2\ 3\ 4\ 5\ 6\ 7\ 8\ 9\ 10\ 11\ 12)
 \end{aligned}$$